AGORA NGELALE

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PROFESSIONAL SUMMARY

Computer Science student with hands-on experience in industrial automation, laboratory research, embedded systems, and software development. At Oak Ridge National Laboratory, contributed to the Laboratory for Uranium Science Technology Center through PLC/HMI development, VFD integration, and research into the uranium life cycle and nuclear nonproliferation. Experienced with Siemens PLCs, HMI/SCADA, VFDs, industrial networks, sensors, motors, system testing, commissioning, and technical documentation.

TECHNICAL SKILLS

Programming: C++, Python, JavaScript, SQL

Software & Web: React, Node.js, Express.js, MongoDB, REST APIs, HTML/CSS, Git/GitHub, Agile/Scrum, SDLC

Embedded Systems: Arduino, Microcontrollers, Hardware Interrupts, Sensors, Analog/Digital I/O, Hardware-Software Integration

Automation & Controls: Siemens S7-1200/S7-1500, Ladder Logic, TIA Portal, HMI/SCADA, VFDs, Servo Motors, Actuators, Control Panels

Industrial Communications: PROFINET, EtherNet/IP, Modbus

Engineering & Test: System Integration, Equipment Bring-Up, Commissioning, Calibration, Validation, Functional/System Testing, Root Cause Analysis, Preventive Maintenance

Research & Documentation: Nuclear Nonproliferation, Uranium Life Cycle, Technical Research, Technical Presentations, Electrical Schematics, Wiring Diagrams, Test Documentation, Microsoft Office

ENGINEERING EXPERIENCE

Oak Ridge National Laboratory (ORNL) **Jun 2026 - Aug 2026**

Research Student Intern | Automation, PLC, SCADA & HMI | Oak Ridge, TN

- Developed, integrated, tested, and debugged automation software for laboratory systems using pumps, automated valves, pressure sensors, motors, and analog/digital I/O.
- Programmed Siemens S7-1200/S7-1500 PLC logic for automated sequences, timing, interlocks, device-state handling, and sensor-driven process behavior.
- Integrated and tested VFD-controlled equipment in the laboratory and designed reusable HMI faceplates for remote VFD control, status monitoring, and operator interaction.
- Developed HMI/SCADA screens, process controls, status indicators, and diagnostic tools; supported equipment bring-up, calibration, commissioning, and validation.
- Troubleshot control logic, communications, sensor behavior, and hardware/software integration issues using PROFINET, EtherNet/IP, and Modbus.
- Worked directly with engineers, researchers, and equipment users to translate requirements into reliable controls and document system behavior, test results, and troubleshooting procedures.

RESEARCH EXPERIENCE

Oak Ridge National Laboratory (ORNL) **Summer 2026**

Uranium Science & Nuclear Nonproliferation Research | Oak Ridge, TN

- Conducted research into the uranium life cycle and nuclear nonproliferation, building technical understanding of uranium processing and safeguards-related challenges.
- Contributed to the development of the Laboratory for Uranium Science Technology Center, supporting a laboratory environment intended for uranium science, technology, and process research.
- Evaluated how PLC-based automation can streamline uranium-processing workflows while improving process safety, repeatability, monitoring, and data gathering.
- Presented findings on the use of PLCs in uranium processing, connecting automation and controls concepts with safer operation and improved collection of process data.

SOFTWARE & EMBEDDED PROJECTS

Social Media Web Application | React, Node.js, Express.js, MongoDB, JavaScript

- Built a full-stack application with user authentication, social features, file uploads, REST APIs, and persistent database storage.

Motion-Activated Embedded Control System | C++, Arduino Uno, PIR Sensors

- Developed embedded C++ software interfacing with sensor hardware and implemented hardware interrupts for responsive real-time event handling.

Automated Gas-Sampling Control System | Siemens S7-1200/S7-1500, TIA Portal, HMI/SCADA

- Designed and programmed PLC/HMI controls for an automated laboratory gas-sampling system coordinating vacuum/turbo pumps, automated valves, pressure sensing, timing sequences, and safety interlocks.

EDUCATION

University of California, Riverside - B.S. Computer Science **Expected Dec. 2026**

GPA: 3.2 | Relevant Coursework: Data Structures & Algorithms, Embedded Systems, Operating Systems, Computer Architecture, Software Construction, Databases, Circuits, Logic Design, Compiler Design

Irvine Valley College - A.S. Mathematics and Computer Science **May 2022**

GPA: 3.7